

Handouts-1

Complex Variables

Problem-1: Express the complex number, $\frac{1+i}{1-i}$ in $A + iB$ form, then find its modulus and argument.

Solution:

Step 1: Multiply by the Conjugate

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{(1+i)(1+i)}{(1-i)(1+i)}$$

Compute the denominator:

$$(1-i)(1+i) = 1^2 - i^2 = 1 - (-1) = 1 + 1 = 2$$

Compute the numerator:

$$\begin{aligned}(1+i)(1+i) &= 1(1) + 1(i) + i(1) + i(i) \\ &= 1 + i + i + i^2 \\ &= 1 + 2i - 1 = 2i\end{aligned}$$

Thus,

$$\frac{1+i}{1-i} = \frac{2i}{2} = i$$

So in $A + iB$ form:

$$0 + i1$$

Step 2: Find the Modulus

The modulus of a complex number $z = A + iB$ is:

$$|z| = \sqrt{A^2 + B^2}$$

Since $A = 0$ and $B = 1$:

$$|z| = \sqrt{0^2 + 1^2} = \sqrt{1} = 1$$

Step 3: Find the Argument

The argument θ of $z = A + iB$ is given by:

$$\theta = \tan^{-1} \left(\frac{B}{A} \right)$$

Since $A = 0$ and $B = 1$, we have:

$$\theta = \tan^{-1} \left(\frac{1}{0} \right)$$

Since the complex number is purely imaginary and lies on the positive imaginary axis, the argument is:

$$\theta = \frac{\pi}{2}$$

Final Answer

- In $A + iB$ form: $0 + i1$
- Modulus: 1
- Argument: $\frac{\pi}{2}$

Problem-2: Express the complex number, $\frac{3+2i}{4-i}$ in $A + iB$ form, then find its modulus and argument.

Solution:

Step 1: Multiply by the Conjugate

Multiply numerator and denominator by the conjugate of the denominator, $4 + i$:

$$\frac{3 + 2i}{4 - i} \times \frac{4 + i}{4 + i} = \frac{(3 + 2i)(4 + i)}{(4 - i)(4 + i)}$$

Step 2: Compute the Denominator

$$(4 - i)(4 + i) = 4^2 - i^2 = 16 + 1 = 17$$

Step 3: Compute the Numerator

Expanding $(3 + 2i)(4 + i)$:

$$\begin{aligned} &= 3(4) + 3(i) + 2i(4) + 2i(i) \\ &= 12 + 3i + 8i + 2i^2 \\ &= 12 + 11i + 2(-1) \\ &= 12 - 2 + 11i = 10 + 11i \end{aligned}$$

Step 4: Express in $A + iB$ Form

$$\frac{10 + 11i}{17} = \frac{10}{17} + i\frac{11}{17}$$

Thus, $A = \frac{10}{17}$, $B = \frac{11}{17}$.

Step 5: Find the Modulus

$$\begin{aligned}|z| &= \sqrt{A^2 + B^2} = \sqrt{\left(\frac{10}{17}\right)^2 + \left(\frac{11}{17}\right)^2} \\&= \sqrt{\frac{100}{289} + \frac{121}{289}} \\&= \sqrt{\frac{221}{289}} = \frac{\sqrt{221}}{17}\end{aligned}$$

Step 6: Find the Argument

$$\theta = \tan^{-1} \left(\frac{B}{A} \right) = \tan^{-1} \left(\frac{11}{10} \right)$$

Thus, the **modulus** is $\frac{\sqrt{221}}{17}$ and the **argument** is $\tan^{-1} \left(\frac{11}{10} \right)$.

Home Work:

Simplify and express in $A + iB$ form, then find the modulus and argument:

1. $\frac{1+2i}{3+4i}$
2. $\frac{4-3i}{5+2i}$
3. $\frac{6+i}{2-i}$
4. $\frac{7-5i}{1+i}$
5. $\frac{3+4i}{2-3i}$
6. $\frac{5+6i}{4-i}$
7. $\frac{2-i}{3+2i}$
8. $\frac{8+3i}{1-4i}$
9. $\frac{9-2i}{2+5i}$
10. $\frac{10+i}{4-2i}$

Find the modulus and argument for the following complex numbers:

1. $\left(\frac{3+4i}{2-i}\right)^2$
2. $\frac{(2-3i)(4+i)}{(1+i)(3-2i)}$
3. $\frac{(5-2i)^3}{(2+i)^2}$
4. $\frac{(1-i)(3+2i)(4-i)}{(2+3i)(1-2i)}$

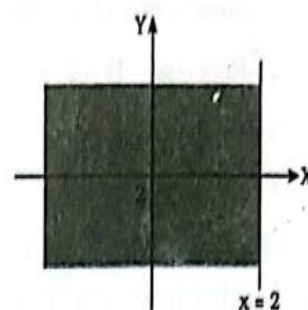
Describe geometrically the region of the following:

- (i) $|z - 4| > |z|$
- (ii) $|z - i| = |z + i|$
- (iii) $\text{Im}(z) > 1$
- (iv) $\text{Re}(z - 1) = 2$
- (v) $|z + 3i| > 4$
- (vi) $|z| > 4$
- (vii) $|z - 2 + i| \leq 4$
- (viii) $|2z + 3| > 4$
- (ix) $\left| \frac{z-3}{z+3} \right| = 3$
- (x) $\left| \frac{z-3}{z+3} \right| > 3$
- (xi) $\left| \frac{z-3}{z+3} \right| < 3$

Solution: (i) Let $z = x + iy$.

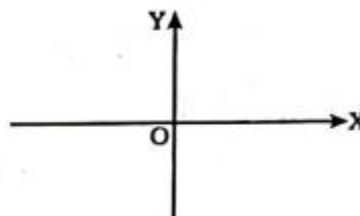
Then $|z - 4| > |z|$ gives

$$\begin{aligned}
 |x + iy - 4| &> |x + iy| \\
 \sqrt{(x-4)^2 + y^2} &> \sqrt{x^2 + y^2} \\
 x^2 - 8x + 16 + y^2 &> x^2 + y^2 \\
 -8x + 16 &> 0 \\
 8x &< 16 \Rightarrow x < 2
 \end{aligned}$$



$\therefore$ The region is the set of all points (x, y) such that $x < 2$. That is, the set of all points (x, y) on the left-hand side of the straight line $x = 2$.

$$\begin{aligned}
 \text{(ii)} \quad |z - i| &= |z + i| \\
 \Rightarrow |x + iy - i| &= |x + iy + i| \\
 \Rightarrow |x + i(y - 1)| &= |x + i(y + 1)| \\
 \Rightarrow \sqrt{x^2 + (y - 1)^2} &= \sqrt{x^2 + (y + 1)^2} \\
 \Rightarrow x^2 + y^2 - 2y + 1 &= x^2 + y^2 + 2y + 1 \\
 \Rightarrow -2y &= 2y \\
 \Rightarrow 4y &= 0 \Rightarrow y = 0, \text{ which is the equation of real axis (x - axis).}
 \end{aligned}$$

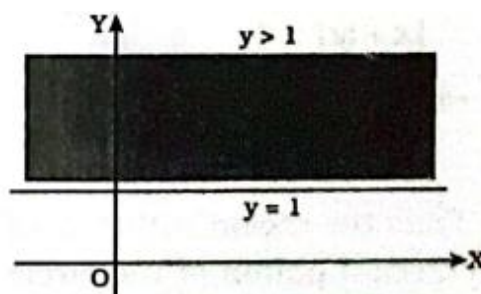


Thus the region is the set of all points lie on the real axis.

(iii) $\text{Im}(z) > 1$

$\Rightarrow \text{Im}(x + iy) > 1 \Rightarrow y > 1$

The region is the set of all points (x, y) such that $y > 1$, that is, the set of all points (x, y) lie in the upper of the line $y = 1$.



(iv) $\text{Re}(\bar{z} - 1) = 2$

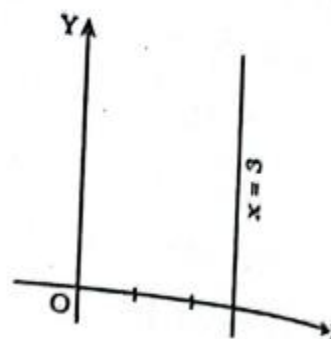
$\Rightarrow \text{Re}(\overline{x + iy} - 1) = 2$, where $z = x + iy$

$\Rightarrow \text{Re}(x - iy - 1) = 2$

$\Rightarrow x - 1 = 2$

$\Rightarrow x = 3$

Which is the equation of a straight line parallel to y-axis. Thus the region is the set of all points $(3, y)$, where $y \in \mathbb{R}$.



(v) Let $z = x + iy$. Then $|z + 3i| > 4$ gives

$|x + iy + 3i| > 4$

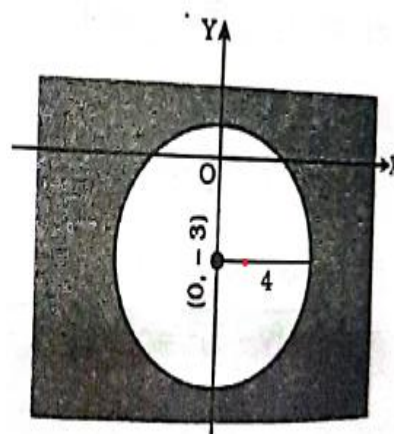
$|x + i(y + 3)| > 4$

$\sqrt{x^2 + (y + 3)^2} > 4$

$x^2 + (y + 3)^2 > 4^2$

$x^2 + (y + 3)^2 = 4^2$ is the equation of a circle whose centre is $(0, -3)$ and radius is 4.

Thus, the region is the set of all external points of the circle whose centre is $(0, -3)$ and radius is



(vii) Let $z = x + iy$. Then $|z - 2 + i| \leq 1$ gives

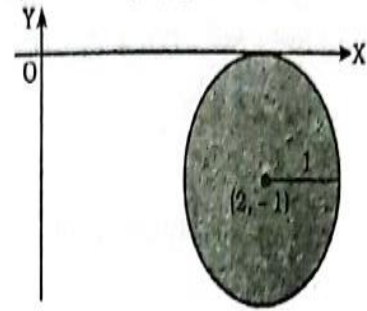
$$|x + iy - 2 + 1| \leq 1$$

$$|(x - 2) + i(y + 1)| \leq 1$$

$$\sqrt{(x - 2)^2 + (y + 1)^2} \leq 1$$

$$(x - 2)^2 + (y + 1)^2 \leq 1$$

Thus, the region is the set of all internal points including the boundary points of a circle whose centre is $(2, -1)$ and radius is 1.



Analytic function: A complex function $f(z)$ is said to be analytic at a point z_0 if its derivative exists not only at z_0 but also at each point z in some neighborhood of z_0 .

Here are some examples of analytic functions:

1. Polynomial Functions

- Any polynomial function of z is analytic everywhere in the complex plane.
- Example:

$$f(z) = z^2 + 3z + 5$$

2. Exponential Function

- The function e^z is analytic everywhere.
- Example:

$$f(z) = e^z$$

where $e^z = e^x(\cos y + i \sin y)$ (Euler's formula).

3. Trigonometric Functions

- The sine and cosine functions are analytic everywhere.
- Example:

$$f(z) = \sin z, \quad g(z) = \cos z$$

where $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$ and $\cos z = \frac{e^{iz} + e^{-iz}}{2}$.

4. Logarithm Function

- The complex logarithm $\log z$ is analytic except at branch points (where $z = 0$).
- Example:

$$f(z) = \log z$$

is analytic in the domain excluding the negative real axis (branch cut).

5. Rational Functions

- A function of the form $\frac{P(z)}{Q(z)}$, where P and Q are polynomials, is analytic except at points where $Q(z) = 0$.
- Example:

$$f(z) = \frac{z}{z-1}$$

is analytic everywhere except at $z = 1$.

Cauchy-Riemann PDE: If $W = f(z) = u(x, y) + iv(x, y)$, then the Cauchy-Riemann PDE are

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Necessary conditions for $f(z)$ to be analytic:

The necessary condition for $W = f(z) = u(x, y) + iv(x, y)$ to be analytic at a point $z = x + iy$ of its domain is that the four partial derivatives

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y}, \quad \frac{\partial v}{\partial x}, \quad \text{and} \quad \frac{\partial v}{\partial y}$$

should exist and satisfy the Cauchy-Riemann PDE:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Problem-1:

* Prove that the function $f(z) = z^2 + 5iz + 3 - i$ satisfies the Cauchy-Riemann equations.

Soln: Given that $f(z) = z^2 + 5iz + 3 - i$

$$\begin{aligned}\Rightarrow u + iv &= (x + iy)^2 + 5i(x + iy) + 3 - i \\ &= x^2 + 2ixy - y^2 + 5ix - 5y + 3 - i \\ &= (x^2 - y^2 - 5y + 3) + i(2xy + 5x - 1)\end{aligned}$$

Equating real and imaginary parts, we get

$$\begin{aligned}u &= x^2 - y^2 - 5y + 3 \quad \text{and} \quad v = 2xy + 5x - 1 \\ \therefore \frac{\partial u}{\partial x} &= 2x, \quad \frac{\partial u}{\partial y} = -2y - 5, \quad \frac{\partial v}{\partial x} = 2y + 5, \quad \text{and} \quad \frac{\partial v}{\partial y} = 2x\end{aligned}$$

$\therefore$ The Cauchy-Riemann equations have been satisfied.

Problem-2: Verify $f(z) = e^z$ satisfies the Cauchy-Riemann equations.

Solution:

Given function:

$$f(z) = e^z = e^x(\cos y + i \sin y)$$

Expanding into real and imaginary parts:

$$u(x, y) = e^x \cos y, \quad v(x, y) = e^x \sin y$$

Now, compute the partial derivatives:

$$\begin{aligned}\frac{\partial u}{\partial x} &= e^x \cos y, & \frac{\partial u}{\partial y} &= -e^x \sin y \\ \frac{\partial v}{\partial x} &= e^x \sin y, & \frac{\partial v}{\partial y} &= e^x \cos y\end{aligned}$$

Checking the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (\checkmark \text{ Satisfied})$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (\checkmark \text{ Satisfied})$$

Since both equations hold, $f(z) = e^z$ is analytic.

Problem-3:

Verify $f(z) = x^2 - y^2 + i(2xy)$ satisfies the Cauchy-Riemann equations

Solution:

Given function:

$$f(z) = x^2 - y^2 + i(2xy)$$

So, the real and imaginary parts are:

$$u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy$$

Now, compute the partial derivatives:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y, \quad \frac{\partial v}{\partial y} = 2x$$

Checking the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad (\checkmark \text{ Satisfied})$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (\checkmark \text{ Satisfied})$$

Since both equations hold, $f(z) = x^2 - y^2 + i(2xy)$ is analytic.

Verify $f(z) = z^2 + 4z + 6$ satisfies the Cauchy-Riemann equations

Solution:

Given function:

$$f(z) = z^2 + 4z + 6$$

Writing in terms of x and y :

$$f(z) = (x + iy)^2 + 4(x + iy) + 6$$

Expanding:

$$\begin{aligned} &= x^2 + 2ixy - y^2 + 4x + 4iy + 6 \\ &= (x^2 - y^2 + 4x + 6) + i(2xy + 4y) \end{aligned}$$

So, the real and imaginary parts are:

$$u(x, y) = x^2 - y^2 + 4x + 6, \quad v(x, y) = 2xy + 4y$$

Now, compute the partial derivatives:

$$\frac{\partial u}{\partial x} = 2x + 4, \quad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y, \quad \frac{\partial v}{\partial y} = 2x + 4$$

Checking the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \Rightarrow (2x + 4 = 2x + 4) \quad (\checkmark \text{ Satisfied})$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \Rightarrow (-2y = -2y) \quad (\checkmark \text{ Satisfied})$$

Since both equations hold, $f(z) = z^2 + 4z + 6$ is analytic.

Exercise Set

1. Verify whether the function $f(z) = z^3$ is analytic.

(Hint: Express $z = x + iy$, expand $f(z) = (x + iy)^3$, separate real and imaginary parts, and check Cauchy-Riemann equations.)

2. Check if $f(z) = e^x(\cos y + i \sin y)$ is analytic.

(Hint: This is e^z , use the Cauchy-Riemann conditions.)

3. Determine whether $f(z) = x^3 - 3xy^2 + i(3x^2y - y^3)$ satisfies the Cauchy-Riemann equations.

5. Is the function $f(z) = x^2 + y^2 + i(2xy)$ analytic?

6. Check the analyticity of $f(z) = \frac{1}{z}$, where $z \neq 0$.

(Hint: Write it in terms of x, y and check the derivatives.)

7. Verify whether $f(z) = \frac{x-iy}{x+iy}$ satisfies the Cauchy-Riemann equations.

9. Check if $f(z) = e^{-y}(\cos x + i \sin x)$ is analytic.

10. Verify analyticity of $f(z) = x^2 - y^2 + 2i(xy + 1)$.

Harmonic Function

A harmonic function is a twice continuously differentiable real-valued function $u(x, y)$ that satisfies Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

for all points (x, y) in a given domain.

Examples of Harmonic Functions

1. $u(x, y) = x^2 - y^2$
2. $u(x, y) = e^x \cos y$
3. $v(x, y) = e^x \sin y$
4. $u(x, y) = \ln \sqrt{x^2 + y^2}$

Problem-1: i) Show that the function $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is harmonic and ii) find its harmonic conjugate v , where $f(z) = u + iv$ is analytic.

Solution:

(i) Show that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is harmonic

A function $u(x, y)$ is harmonic if it satisfies Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Step 1: Compute $\frac{\partial^2 u}{\partial x^2}$

Given:

$$u = x^3 + 6x^2y - 3xy^2 - 2y^3$$

First derivative with respect to x :

$$\frac{\partial u}{\partial x} = 3x^2 + 12xy - 3y^2.$$

Second derivative with respect to x :

$$\frac{\partial^2 u}{\partial x^2} = 6x + 12y.$$

Step 2: Compute $\frac{\partial^2 u}{\partial y^2}$

First derivative with respect to y :

$$\frac{\partial u}{\partial y} = 6x^2 - 6xy - 6y^2.$$

Second derivative with respect to y :

$$\frac{\partial^2 u}{\partial y^2} = -6x - 12y.$$

Step 3: Check Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = (6x + 12y) + (-6x - 12y) = 0.$$

Since Laplace's equation is satisfied, $u(x, y)$ is a harmonic function. 

(ii) Find the harmonic conjugate $v(x, y)$

A function $v(x, y)$ is a harmonic conjugate of $u(x, y)$ if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

From earlier calculations:

$$\frac{\partial u}{\partial x} = 3x^2 + 12xy - 3y^2.$$

$$\frac{\partial u}{\partial y} = 6x^2 - 6xy - 6y^2.$$

Using the Cauchy-Riemann equations:

$$\frac{\partial v}{\partial y} = 3x^2 + 12xy - 3y^2.$$

$$\frac{\partial v}{\partial x} = -(6x^2 - 6xy - 6y^2) = -6x^2 + 6xy + 6y^2.$$

Step 1: Integrate $\frac{\partial v}{\partial y}$

$$\frac{\partial v}{\partial y} = 3x^2 + 12xy - 3y^2.$$

Integrating with respect to y :

$$v = 3x^2y + 6xy^2 - y^3 + C(x).$$

(where $C(x)$ is a function of x only)

Step 2: Differentiate with respect to x

$$\frac{\partial v}{\partial x} = 6xy + 6y^2 + C'(x).$$

From the Cauchy-Riemann equation:

$$6xy + 6y^2 + C'(x) = -6x^2 + 6xy + 6y^2.$$

Canceling terms:

$$C'(x) = -6x^2.$$

Integrating:

$$C(x) = -2x^3 + C_0.$$

Thus, the harmonic conjugate is:

$$v = 3x^2y + 6xy^2 - y^3 - 2x^3 + C_0.$$

Final Answer

1. $u(x, y) = x^3 + 6x^2y - 3xy^2 - 2y^3$ is harmonic.
2. The harmonic conjugate is:

$$v(x, y) = 3x^2y + 6xy^2 - y^3 - 2x^3 + C_0.$$

Since the constant C_0 does not affect analyticity, it can be ignored.

Problem 2:

- (i) Show that the function

$$u = x^3 - 3xy^2 + 2x + 4y$$

is **harmonic** (i.e., it satisfies Laplace's equation).

- (ii) Find its harmonic conjugate $v(x, y)$, where $f(z) = u + iv$ is analytic.

Solution:

Step 1: Check if $u(x, y)$ is harmonic

A function $u(x, y)$ is **harmonic** if it satisfies **Laplace's equation**:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

First, compute the first-order partial derivatives:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(x^3 - 3xy^2 + 2x + 4y) \\ &= 3x^2 - 3y^2 + 2.\end{aligned}$$

$$\begin{aligned}\frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(x^3 - 3xy^2 + 2x + 4y) \\ &= -6xy + 4.\end{aligned}$$

Now, compute the second-order partial derivatives:

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x}(3x^2 - 3y^2 + 2) \\ &= 6x.\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y}(-6xy + 4) \\ &= -6x.\end{aligned}$$

Now, check Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6x + (-6x) = 0.$$

Since Laplace's equation is satisfied, $u(x, y)$ is **harmonic**.

Step 2: Find the harmonic conjugate $v(x, y)$

A function $v(x, y)$ is the **harmonic conjugate** of $u(x, y)$ if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

We already found:

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 + 2.$$

$$\frac{\partial u}{\partial y} = -6xy + 4.$$

From Cauchy-Riemann equations:

$$\frac{\partial v}{\partial y} = 3x^2 - 3y^2 + 2. \quad (\text{Equation 1})$$

$$\frac{\partial v}{\partial x} = -(-6xy + 4) = 6xy - 4. \quad (\text{Equation 2})$$

To find $v(x, y)$, integrate Equation 2 with respect to x :

$$\begin{aligned} v(x, y) &= \int (6xy - 4) dx. \\ &= 3x^2y - 4x + g(y), \end{aligned}$$

where $g(y)$ is an arbitrary function of y .

Now, differentiate $v(x, y)$ with respect to y and compare with Equation 1:

$$\frac{\partial v}{\partial y} = 3x^2 + g'(y).$$

Equating this with Equation 1:

$$3x^2 + g'(y) = 3x^2 - 3y^2 + 2.$$

$$g'(y) = -3y^2 + 2.$$

Integrating with respect to y :

$$g(y) = -y^3 + 2y + C,$$

where C is an arbitrary constant.

Thus, the harmonic conjugate is:

$$v(x, y) = 3x^2y - 4x - y^3 + 2y + C.$$

Final Answer:

(i) $u(x, y) = x^3 - 3xy^2 + 2x + 4y$ is harmonic.

(ii) The harmonic conjugate is:

$$v(x, y) = 3x^2y - 4x - y^3 + 2y + C.$$

3.7. (a) Prove that $u = e^{-x}(x \sin y - y \cos y)$ is harmonic.

(b) Find v such that $f(z) = u + iv$ is analytic.

Solution

$$(a) \quad \frac{\partial u}{\partial x} = (e^{-x})(\sin y) + (-e^{-x})(x \sin y - y \cos y) = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}(e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y) = -2e^{-x} \sin y + xe^{-x} \sin y - ye^{-x} \cos y \quad (1)$$

$$\frac{\partial u}{\partial y} = e^{-x}(x \cos y + y \sin y - \cos y) = xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}(xe^{-x} \cos y + ye^{-x} \sin y - e^{-x} \cos y) = -xe^{-x} \sin y + 2e^{-x} \sin y + ye^{-x} \cos y \quad (2)$$

Adding (1) and (2) yields $(\partial^2 u / \partial x^2) + (\partial^2 u / \partial y^2) = 0$ and u is harmonic.

(b) From the Cauchy–Riemann equations,

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = e^{-x} \sin y - xe^{-x} \sin y + ye^{-x} \cos y \quad (3)$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = e^{-x} \cos y - xe^{-x} \cos y - ye^{-x} \sin y \quad (4)$$

Integrate (3) with respect to y , keeping x constant. Then

$$\begin{aligned} v &= -e^{-x} \cos y + xe^{-x} \cos y + e^{-x}(y \sin y + \cos y) + F(x) \\ &= ye^{-x} \sin y + xe^{-x} \cos y + F(x) \end{aligned} \quad (5)$$

where $F(x)$ is an arbitrary real function of x .

Substitute (5) into (4) and obtain

$$-ye^{-x} \sin y - xe^{-x} \cos y + e^{-x} \cos y + F'(x) = -ye^{-x} \sin y - xe^{-x} \cos y - ye^{-x} \sin y$$

or $F'(x) = 0$ and $F(x) = c$, a constant. Then, from (5),

$$v = e^{-x}(y \sin y + x \cos y) + c$$

Exercise Set on Harmonic Functions and Their Conjugates

Problem 1

Show that the function

$$u = x^2 - y^2 + 4x - 3y$$

is harmonic, and find its harmonic conjugate $v(x, y)$.

Problem 2

Show that the function

$$u = e^x \cos y$$

is harmonic, and find its harmonic conjugate $v(x, y)$.

Problem 3

Verify that the function

$$u = \ln(x^2 + y^2)$$

is harmonic, and find its harmonic conjugate $v(x, y)$.

Problem 4

Show that the function

$$u = \sin x \sin y$$

is harmonic, and find its harmonic conjugate $v(x, y)$.

Problem 5

Verify whether the function

$$u = x^3 - 3xy^2$$

is harmonic, and if so, find its harmonic conjugate $v(x, y)$.

Problem 6

Show that the function

$$u = e^x \sin y$$

is harmonic, and determine its harmonic conjugate $v(x, y)$.

Problem 7

Prove that

$$u = \frac{x}{x^2 + y^2}$$

is **harmonic**, and find its harmonic conjugate $v(x, y)$.

Problem 8

Check whether the function

$$u = \cos x \cos y$$

is **harmonic**, and if so, determine its harmonic conjugate $v(x, y)$.

Complex Integration

Show that

$$\int_{(0,1)}^{(2,5)} (3x + y)dx + (2y - x)dy = 32$$

along the straight line joining the points $(0, 1)$ and $(2, 5)$.

Solution

The equation of the straight line joining the points $(0, 1)$ and $(2, 5)$ is:

$$\frac{y - 1}{1 - 5} = \frac{x - 0}{0 - 2}$$

$$\frac{y - 1}{-4} = \frac{x}{-2}$$

$$y - 1 = 2x$$

$$y = 2x + 1$$

$$\therefore dy = 2dx$$

Now, evaluating the given integral:

$$\int_{(0,1)}^{(2,5)} (3x + y)dx + (2y - x)dy$$

Substituting $y = 2x + 1$ and $dy = 2dx$:

$$\begin{aligned} & \int_0^2 (3x + 2x + 1)dx + \int_0^2 (2(2x + 1) - x)2dx \\ &= \int_0^2 (5x + 1)dx + \int_0^2 (4x + 2 - x)2dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^2 (5x + 1)dx + \int_0^2 (3x + 2)2dx \\
&= \int_0^2 (5x + 1 + 6x + 4)dx \\
&= \int_0^2 (11x + 5)dx \\
&= \left[\frac{11x^2}{2} + 5x \right]_0^2 \\
&= \left[\frac{11(2)^2}{2} + 5(2) \right] - \left[\frac{11(0)^2}{2} + 5(0) \right] \\
&= \left[\frac{44}{2} + 10 \right] - 0 \\
&= (22 + 10) = 32 \\
&\therefore \boxed{32}
\end{aligned}$$

Find the value of the integral

$$\int_C (x + y)dx + x^2ydy$$

1. Along $y = x^2$ having $(0, 0)$ and $(3, 9)$ as end points.
2. Along $y = 3x$ between the same points.

Solution:

(i) Along $y = x^2$:

Since $y = x^2$, we have $dy = 2x dx$.

$$\begin{aligned} & \int_0^3 (x + x^2) dx + x^2 \cdot x \cdot 2x dx \\ &= \int_0^3 (x + x^2 + 2x^5) dx \\ &= \left[\frac{x^2}{2} + \frac{x^3}{3} + \frac{2x^6}{6} \right]_0^3 \\ &= \frac{256}{2} \end{aligned}$$

(ii) Along $y = 3x$:

Since $y = 3x$, we have $dy = 3dx$.

$$\begin{aligned} & \int_0^3 (x + 3x) dx + x^2 \cdot 3x(3dx) \\ &= \int_0^3 4x dx + 9 \int_0^3 x^3 dx \\ &= \left[2x^2 + \frac{9}{4}x^4 \right]_0^3 \\ &= \frac{200}{4} \end{aligned}$$

Evaluate

$$\int_C (2y + x^2)dx + (3x - y)dy$$

Along:

1. The straight-line path from $(0, 3)$ to $(2, 3)$, then from $(2, 3)$ to $(2, 4)$.
2. The straight-line path from $(0, 3)$ to $(2, 4)$.

Solution:

(i) Path from $(0, 3)$ to $(2, 3)$, then from $(2, 3)$ to $(2, 4)$

Step 1: From $(0, 3)$ to $(2, 3)$

- Along this path, $y = 3$, so $dy = 0$.
- The integral simplifies to:

$$\begin{aligned}\int_0^2 (2(3) + x^2)dx \\&= \int_0^2 (6 + x^2)dx \\&= \left[6x + \frac{x^3}{3} \right]_0^2 \\&= \left(6(2) + \frac{2^3}{3} \right) - (0) \\&= 12 + \frac{8}{3} = \frac{36}{3} + \frac{8}{3} = \frac{44}{3}\end{aligned}$$

Step 2: From $(2, 3)$ to $(2, 4)$

- Along this path, $x = 2$, so $dx = 0$.
- The integral simplifies to:

$$\begin{aligned}
& \int_3^4 (3(2) - y) dy \\
&= \int_3^4 (6 - y) dy \\
&= \left[6y - \frac{y^2}{2} \right]_3^4 \\
&= \left(6(4) - \frac{4^2}{2} \right) - \left(6(3) - \frac{3^2}{2} \right) \\
&= (24 - 8) - (18 - 4.5) \\
&= 16 - 13.5 = 2.5
\end{aligned}$$

Total integral value for (i):

$$\frac{44}{3} + 2.5 = \frac{44}{3} + \frac{5}{2} = \frac{88}{6} + \frac{15}{6} = \frac{103}{6}$$

(ii) Path along a straight line from (0, 3) to (2, 4)

Step 1: Equation of the straight line

The equation of the straight line passing through (0, 3) and (2, 4) is obtained using:

$$\frac{y - 3}{4 - 3} = \frac{x - 0}{2 - 0}$$

$$y - 3 = x$$

$$y = x + 3 \Rightarrow dy = dx$$

Step 2: Substituting into the integral

$$\begin{aligned} & \int_0^2 (2(x+3) + x^2) dx + \int_0^2 (3x - (x+3)) dx \\ &= \int_0^2 (2x + 6 + x^2) dx + \int_0^2 (3x - x - 3) dx \\ &= \int_0^2 (x^2 + 2x + 6) dx + \int_0^2 (2x - 3) dx \end{aligned}$$

Step 3: Evaluating the integrals

$$\begin{aligned} & \left[\frac{x^3}{3} + x^2 + 6x \right]_0^2 + [x^2 - 3x]_0^2 \\ &= \left(\frac{2^3}{3} + 2^2 + 6(2) \right) - (0) + (2^2 - 3(2)) - (0) \\ &= \left(\frac{8}{3} + 4 + 12 \right) + (4 - 6) \\ &= \left(\frac{8}{3} + 16 \right) + (-2) \\ &= \frac{8}{3} + 14 = \frac{8}{3} + \frac{42}{3} = \frac{50}{3} \end{aligned}$$

Final Answer:

1. For path (i): $\frac{103}{6}$
2. For path (ii): $\frac{50}{3}$

Exercise Set: Line Integral Problems

1. Evaluate the line integral

$$\int_C (3x + 4y)dx + (2x - y)dy$$

Along:

- (i) The straight-line path from $(1, 2)$ to $(4, 6)$.
- (ii) The piecewise path from $(1, 2)$ to $(4, 2)$, then from $(4, 2)$ to $(4, 6)$.

2. Compute the line integral

$$\int_C (x^2 + y)dx + (xy - 3)dy$$

Along the straight-line path from $(0, 0)$ to $(3, 4)$.

3. Find the value of the integral

$$\int_C (2y + x^2)dx + (5x - y)dy$$

Along:

- (i) $y = x^3$, from $(0, 0)$ to $(2, 8)$.
- (ii) $y = 2x + 1$, from $(0, -1)$ to $(3, 5)$.

4. Evaluate the integral

$$\int_C (4x + 2y)dx + (x - y)dy$$

Along:

- (i) The straight-line segment from $(0, 1)$ to $(3, 4)$.
- (ii) The piecewise path from $(0, 1)$ to $(3, 1)$, then from $(3, 1)$ to $(3, 4)$.

5. Calculate the line integral

$$\int_C (5x^2 + 3y)dx + (4x - 2y^2)dy$$

Along the parametric curve $x = t^2$, $y = 2t$, for t from 0 to 2.

Cauchy's Theorem: If $f(z)$ is analytic at all points inside and on a simple closed curve C then $\oint_C f(z)dz = 0$.

Cauchy's Integral Formula

Let $f(z)$ be analytic inside and on a simple closed curve C . If a is any point inside C , then

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - a} dz$$

where C is traversed in the positive sense, i.e., in the counterclockwise direction.

Generalized Cauchy's Integral Formula (Cauchy's Integral Formula for Derivatives)

If $f(z)$ is analytic inside and on a simple closed curve C , and a is any point inside C , then the n th derivative of $f(z)$ at a is given by:

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - a)^{n+1}} dz$$

where C is traversed in the positive (counterclockwise) direction.

Problems on Cauchy's Integral formula :

1. Evaluate $\oint_C \frac{z^2 - z + 1}{z - 2} dz$; where C is the circle (i) $|z| = 1$,
(ii) $|z| = 3$ and (iii) $|z| = \frac{3}{2}$

Solution:

a) Here we observed that

- i) The closed curve C is a circle of **radius 1** centered at the origin
- ii) The function $f(z) = z^2 - z + 1$ and the point $a = 2$ which lies **outside** the closed curve C
- iii) The function is analytic inside and on the closed curve C: $|z|=1$

So, by **Cauchy's Theorem**, the integral evaluates to:

$$\oint_C \frac{z^2 - z + 1}{z - 2} dz = 0.$$

b) Here we observed that

- i) The closed curve C is a circle of **radius 3** centered at the origin
- ii) The function $f(z) = z^2 - z + 1$ and the point $a = 2$ which lies **inside** the closed curve C
- iii) The function is analytic inside and on the closed curve C: $|z|=3$

Applying **Cauchy's Integral Formula**, we find

$$\oint_C \frac{f(z)}{z - 2} dz = 2\pi i f(2).$$

- Evaluating $f(2)$:

$$f(2) = 2^2 - 2 + 1 = 4 - 2 + 1 = 3.$$

- Therefore, the integral is:

$$\oint_C \frac{z^2 - z + 1}{z - 2} dz = 2\pi i \times 3 = 6\pi i.$$

c) Here we observed that

- i) The closed curve C is a circle of **radius $\frac{3}{2}$** centered at the origin

ii) The function $f(z) = z^2 - z + 1$ and the point $a = 2$ which lies **outside** the closed curve C

iii) The function is analytic inside and on the closed curve C : $|z| = \frac{3}{2}$

So, by **Cauchy's Theorem**, the integral evaluates to:

$$\oint_C \frac{z^2 - z + 1}{z - 2} dz = 0.$$

2. Using Cauchy's integral formula, evaluate

$$\oint_C \frac{z^2 - z + 1}{z - 1} dz$$

where C is the circle, (i) $|z| = 2$ and (ii) $|z| = \frac{1}{2}$.

Solution:

(i) For $|z| = 2$:

- The closed curve C is a **circle of radius 2 centered at the origin**.
- The function given is:

$$f(z) = z^2 - z + 1$$

and the singularity is at $a = 1$, which **lies inside** the closed curve C .

- The function is analytic inside and on the closed curve $|z| = 2$.

Using **Cauchy's Integral Formula**:

$$\oint_C \frac{f(z)}{z - 1} dz = 2\pi i f(1).$$

Calculating $f(1)$:

$$f(1) = 1^2 - 1 + 1 = 1.$$

Thus, the integral evaluates to:

$$\oint_C \frac{z^2 - z + 1}{z - 1} dz = 2\pi i \times 1 = 2\pi i.$$

(ii) For $|z| = \frac{1}{2}$:

- The closed curve C is a circle of radius $\frac{1}{2}$ centered at the origin.
- The function $f(z) = z^2 - z + 1$ and the singularity at $a = 1$, which lies outside the closed curve C .
- Since the function is analytic inside and on the closed curve $|z| = \frac{1}{2}$, by **Cauchy's Theorem**, the integral evaluates to:

$$\oint_C \frac{z^2 - z + 1}{z - 1} dz = 0.$$

Final Answer:

$$\oint_C \frac{z^2 - z + 1}{z - 1} dz = \begin{cases} 2\pi i, & \text{if } |z| = 2, \\ 0, & \text{if } |z| = \frac{1}{2}. \end{cases}$$

Problem-3:

evaluate the contour integral using **Cauchy's Integral Formula**:

$$\frac{1}{2\pi i} \oint_C \frac{e^z}{z - 2} dz.$$

Solution:

(i) For $|z| = 3$:

- The closed curve C is a circle of radius 3 centered at the origin.
- The function given is:

$$f(z) = e^z$$

and the singularity is at $a = 2$, which lies inside the closed curve C .

- The function is analytic inside and on the closed curve $|z| = 3$.

Using **Cauchy's Integral Formula**:

$$\oint_C \frac{f(z)}{z - 2} dz = 2\pi i f(2).$$

Calculating $f(2)$:

$$f(2) = e^2.$$

Thus,

$$\frac{1}{2\pi i} \oint_C \frac{e^z}{z-2} dz = \frac{1}{2\pi i} \times 2\pi i \times e^2 = e^2.$$

(ii) For $|z| = 1$:

- The closed curve C is a **circle of radius 1 centered at the origin**.
- The function $f(z) = e^z$ and the singularity at $a = 2$, which **lies outside** the closed curve C .
- Since the function is analytic inside and on the closed curve $|z| = 1$, by **Cauchy's Theorem**, the integral evaluates to:

$$\oint_C \frac{e^z}{z-2} dz = 0.$$

Thus,

$$\frac{1}{2\pi i} \oint_C \frac{e^z}{z-2} dz = 0.$$

Final Answer:

$$\frac{1}{2\pi i} \oint_C \frac{e^z}{z-2} dz = \begin{cases} e^2, & \text{if } |z| = 3, \\ 0, & \text{if } |z| = 1. \end{cases}$$

Problem 4

Evaluate the contour integral using **Cauchy's Integral Formula**:

$$\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz.$$

where C is the circle $|z| = 5$.

Solution:

- The closed curve C is a circle of radius 5 centered at the origin.
- The function given is:

$$f(z) = \sin 3z$$

and the singularity is at $a = -\frac{\pi}{2}$, which lies inside the closed curve C .

- The function is analytic inside and on the closed curve $|z| = 5$.

Using Cauchy's Integral Formula:

$$\oint_C \frac{f(z)}{z - a} dz = 2\pi i f(a).$$

Substituting $a = -\frac{\pi}{2}$:

$$f\left(-\frac{\pi}{2}\right) = \sin\left(3 \times -\frac{\pi}{2}\right) = \sin\left(-\frac{3\pi}{2}\right).$$

Using the identity $\sin(-x) = -\sin x$:

$$\sin\left(-\frac{3\pi}{2}\right) = -\sin\frac{3\pi}{2} = -(-1) = 1.$$

Thus,

$$\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i \times 1 = 2\pi i.$$

Final Answer:

$$\oint_C \frac{\sin 3z}{z + \frac{\pi}{2}} dz = 2\pi i.$$

Problem 5

Evaluate the contour integral:

$$\oint_C \frac{e^{3z}}{(z-1)(z-2)} dz,$$

where C is the circle $|z| = 4$.

Solution:

Given:

$$\oint_C \frac{e^{3z}}{(z-1)(z-2)} dz$$

We use partial fraction decomposition:

$$\oint_C e^{3z} \left(\frac{1}{z-2} - \frac{1}{z-1} \right) dz$$

Splitting the integral:

$$\oint_C \frac{e^{3z}}{z-2} dz - \oint_C \frac{e^{3z}}{z-1} dz.$$

Since $f(z) = e^{3z}$ and the points $a = 2, 1$ both lie inside C , we apply **Cauchy's Integral Formula**:

$$\oint_C \frac{e^{3z}}{z-a} dz = 2\pi i f(a).$$

Thus,

$$\oint_C \frac{e^{3z}}{z-2} dz - \oint_C \frac{e^{3z}}{z-1} dz$$

Evaluating:

$$= 2\pi i e^{3(2)} - 2\pi i e^{3(1)}$$

$$= 2\pi i(e^6 - e^3).$$

Final Answer:

$$\oint_C \frac{e^{3z}}{(z-1)(z-2)} dz = 2\pi i(e^6 - e^3).$$

Problem 7

Find the value of $\oint_C \frac{\cos \pi z}{z^2-4} dz$; where C denotes the square C whose sides lie along the lines $x = \pm 2, y = \pm 2$, described in the positive sense.

Solution:

$$\text{Given } \oint_C \frac{\cos \pi z}{z^2-4} dz$$

$$= \oint_C \frac{\cos \pi z}{(z+2)(z-2)} dz$$

$$= \frac{1}{4} \oint_C \left(\frac{1}{z-2} - \frac{1}{z+2} \right) \cos \pi z dz$$

$$= \frac{1}{4} \oint_C \frac{\cos \pi z dz}{z-2} - \frac{1}{4} \oint_C \frac{\cos \pi z dz}{z+2}.$$

Here $f(z) = \cos \pi z$ and $a = -2, 2$. The square C is bounded by the lines $x = 2, x = -2, y = 2, y = -2$.

Clearly, $f(z)$ is analytic inside and on the square C , and $a = -2, 2$ both lie inside C , so by Cauchy's integral formula, we get:

$$\frac{1}{4} \oint_C \frac{\cos \pi z dz}{z-2} - \frac{1}{4} \oint_C \frac{\cos \pi z dz}{z+2}$$

$$= \frac{1}{4} \cdot 2\pi i f(a) - \frac{1}{4} \cdot 2\pi i f(a)$$

$$= \frac{1}{2} \pi i \cos \pi(2) - \frac{1}{2} \pi i \cos \pi(-2)$$

$$= \frac{\pi}{2} i (\cos 2\pi - \cos 2\pi)$$

$$= \frac{\pi}{2} i(0)$$

$$= 0.$$

Problem 8

Evaluate

$$\oint_C \frac{e^{iz}}{z^2} dz;$$

where C is the circle $|z| = 1$.

Solution:

Here $f(z) = e^{iz}$ and $a = 0$. Clearly, $f(z)$ is analytic inside and on the circle $|z| = 1$, and $a = 0$ lies inside C .

We know from Cauchy's integral formula for derivatives:

$$\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} \frac{d^n}{dz^n} f(z) \text{ at } z = a \text{ for } n = 0, 1, 2, 3, \dots$$

Thus,

$$\begin{aligned} \oint_C \frac{e^{iz}}{z^2} dz &= \frac{2\pi i}{1!} \frac{d}{dz} (e^{iz}) \text{ at } z = 0. \\ &= 2\pi i \cdot ie^{iz} \text{ at } z = 0. \\ &= 2\pi i^2 e^0. \\ &= -2\pi e^0. \\ &= -2\pi. \end{aligned}$$

Problem 9

Evaluate

$$\oint_C \frac{\sin^2 z}{(z - \frac{\pi}{6})^4} dz;$$

where C is the circle $|z| = 2$.

Solution:

Here $f(z) = \sin^2 z$ and $a = \frac{\pi}{6}$. Clearly, $f(z)$ is analytic inside and on the circle $|z| = 2$, and $a = \frac{\pi}{6}$ lies inside C .

We know from Cauchy's integral formula for derivatives:

$$\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} \frac{d^n}{dz^n} f(z) \text{ at } z = a \text{ for } n = 0, 1, 2, 3, \dots$$

Thus,

$$\begin{aligned} \oint_C \frac{\sin^2 z}{(z - \frac{\pi}{6})^4} dz &= \frac{2\pi i}{3!} \left[\frac{d^3}{dz^3} (\sin^2 z) \right] \text{ at } z = \frac{\pi}{6}. \\ &= \frac{2\pi i}{6} \frac{d^2}{dz^2} [2 \sin z \cos z] \text{ at } z = \frac{\pi}{6}. \\ &= \frac{\pi i}{3} \frac{d^2}{dz^2} [\sin 2z] \text{ at } z = \frac{\pi}{6}. \\ &= \frac{\pi i}{3} \frac{d}{dz} [2 \cos 2z] \text{ at } z = \frac{\pi}{6}. \\ &= \frac{-\pi i}{3} [4 \sin 2z] \text{ at } z = \frac{\pi}{6}. \\ &= -\frac{\pi i}{3} 4 \sin \frac{\pi}{3}. \\ &= -\frac{\pi i}{3} 4 \cdot \frac{\sqrt{3}}{2}. \\ &= -\frac{2\pi i}{\sqrt{3}}. \end{aligned}$$

****Exercise Set: Complex Integration using Cauchy's Integral Formula ****

1. Evaluate

$$\oint_C \frac{e^z}{z-2} dz;$$

where C is the circle $|z| = 3$.

2. Compute

$$\int_C \frac{z^2 + 1}{z^2 - 1} dz;$$

where C is the contour $|z| = 2$.

3. Evaluate

$$\oint_C \frac{e^{iz}}{z^2} dz;$$

where C is the circle $|z| = 1$.

4. Compute

$$\oint_C \frac{z}{(z-1)^3} dz;$$

where C is the circle $|z| = 2$.

5. Find the integral

$$\oint_C \frac{\sin z}{(z - \frac{\pi}{6})^4} dz;$$

where C is the circle $|z| = 2$.

6. Compute

$$\oint_C \frac{e^{z^2}}{(z - 1)^2} dz;$$

where C is the circle $|z| = 1.5$.

7. Evaluate

$$\oint_C \frac{\log(1 + z)}{z^2} dz;$$

where C is the circle $|z| = 1$.

8. Compute

$$\oint_C \frac{z^3}{(z - i)^2} dz;$$

where C is the circle $|z| = 2$.

9. Evaluate

$$\oint_C \frac{\cos z}{z^2 - 4} dz;$$

where C is the square contour with sides along $x = \pm 2, y = \pm 2$, described in the positive sense.

10. Compute

$$\oint_C \frac{z^3 + 1}{z^3 - 1} dz;$$

where C is the circle $|z| = 2$.